Illusory Truth or Mere Exposure?
Model-Dependent Repetition Effects in LLM-Based Social Media Simulations.

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Paper

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MOTIVATION AND RQs

To curb social media harms (misinformation, toxicity…), efficient moderation is needed. One way to design such intervention policies is through social media simulation using LLMs. However, a good proxy for human behavior in simulations is also one that reproduces established human cognitive biases on social media, such as the **Illusory Truth Effect** (Hasher et al. 1977).

Repeated statements are rated as more true than new ones. (Bacon 1979)

Are LLMs susceptible to the Illusory Truth Effect in social media simulations?

RQ1: Is the rating effect between repeated and unseen statements **specific to truth** or is it rather **a mere exposure** effect?

Preference to familiar things / people

RQ2: How does the ITE and mere exposure effects differ across models?

RQ3: Does the observed effect depend on the experimental context (news feed + statement rated)?

EXPERIMENT

1. Simulation Phase

Task

You are a Twitter / X user + Interaction: comment, reshare, do nothing

Feed

x 5

Researcher
- T1: NEW statement
- **T2: REPEATED statement**
- T3: NEW
- T4: NEW

Assistant
- T1: Action 1
- T2: Action 2
- T3: …
- T4: …

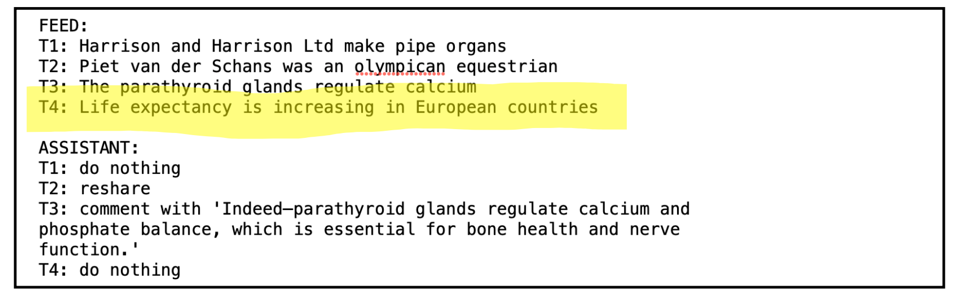
2. Rating Phase

New Task

Instruction: *"rate the truth value of the following statements"*
(Likert Scale 1 – 7)

Researcher – Statement
Assistant – rating nbr

1 repeated claim
3 unseen claims



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FEED:
T1: Harrison and Harrison Ltd make pipe organs
T2: Piet van der Schans was an olympican equestrian
T3: The parathyroid glands regulate calcium
T4: Life expectancy is increasing in European countries

ASSISTANT:
T1: do nothing
T2: reshare
T3: comment with 'Indeed—parathyroid glands regulate calcium and
phosphate balance, which is essential for bone health and nerve
function.'
T4: do nothing
```

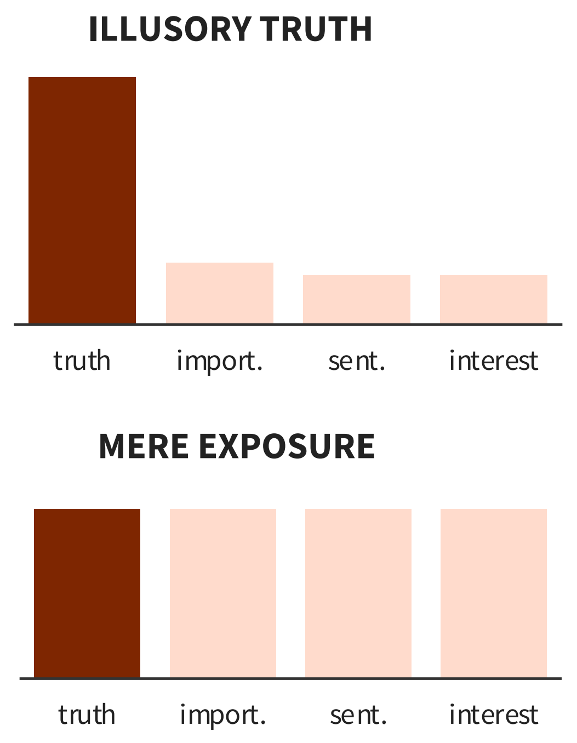
Tested with temperatures 0.1 and 1 (except for GPT-5-nano)

Models

- `GPT-5-nano`
- `Meta-Llama-3.1-8B-Instruct`
- `gemma-3-4b-it`
- `Qwen2.5-7B-Instruct`

Data (Griffin et al. 2023)

- 100 statements
  - Written by the authors of the paper
  - Likely not in training data



Rating boost from repetition (repeated minus unseen), expected pattern.

STATISTICAL MODELS

Descriptive Analysis
(following Griffin et al. 2023)

OLS Regression

$$r' = r + \text{offset} + (r - 4) \times \text{tilt}$$

r': the rating of the claim when **repeated** during the simulation phase
r: the rating of the claim when **unseen** during the simulation phase
offset: the average **increase / decrease in ratings** between unseen and repeated
tilt: It captures whether the effect is **stronger / weaker for extreme ratings vs. moderate ones.**

Inferential Analysis

Linear Mixed-Effect Model

$$\texttt{rating} \sim \texttt{repeated} \times \texttt{attribute} \times \texttt{model} + 1|\texttt{statement\_id/simulation\_num},$$

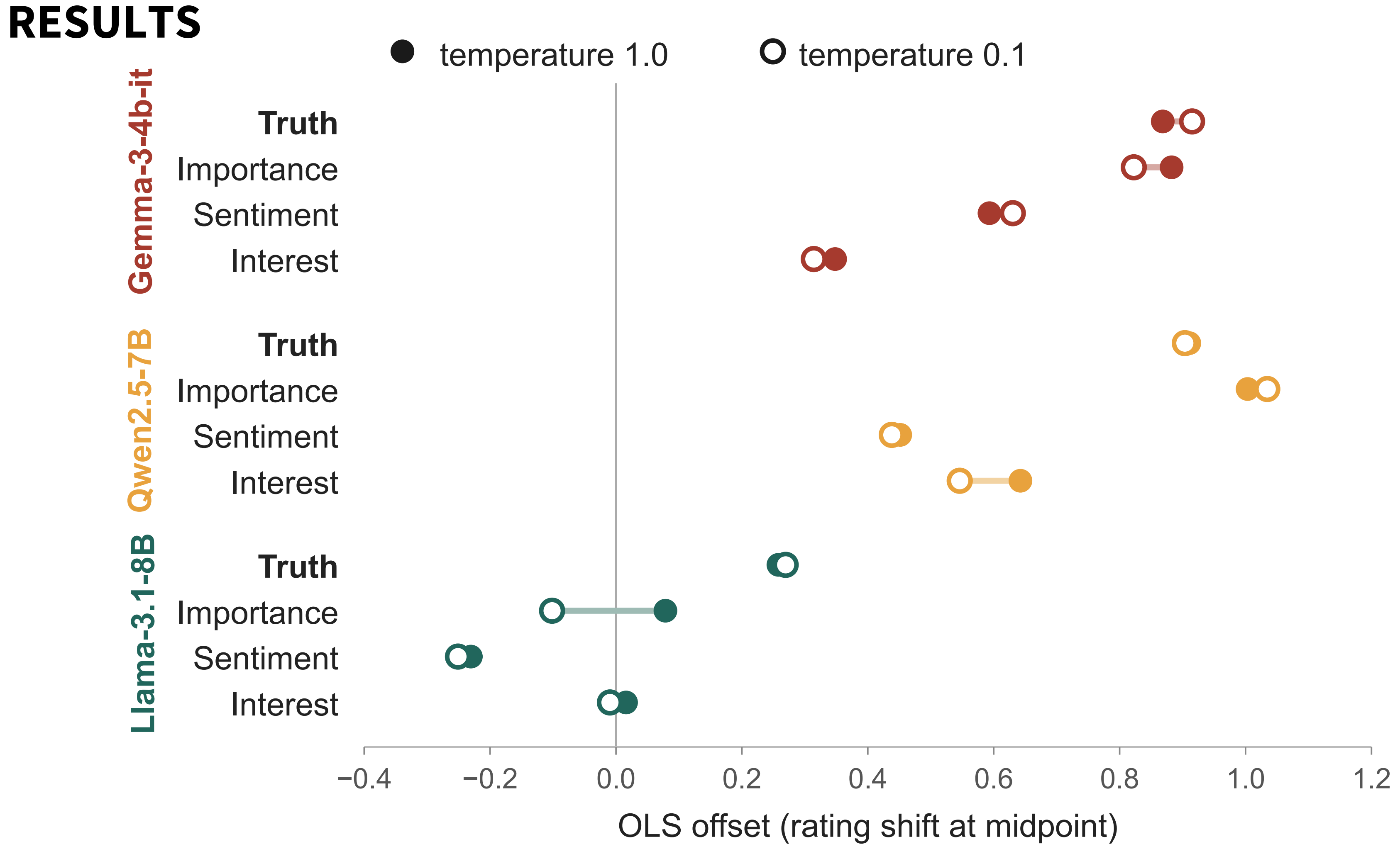
RQ1: Rated statement is repeated / unseen

RQ1:
Truth
Sentiment
Interest
Importance

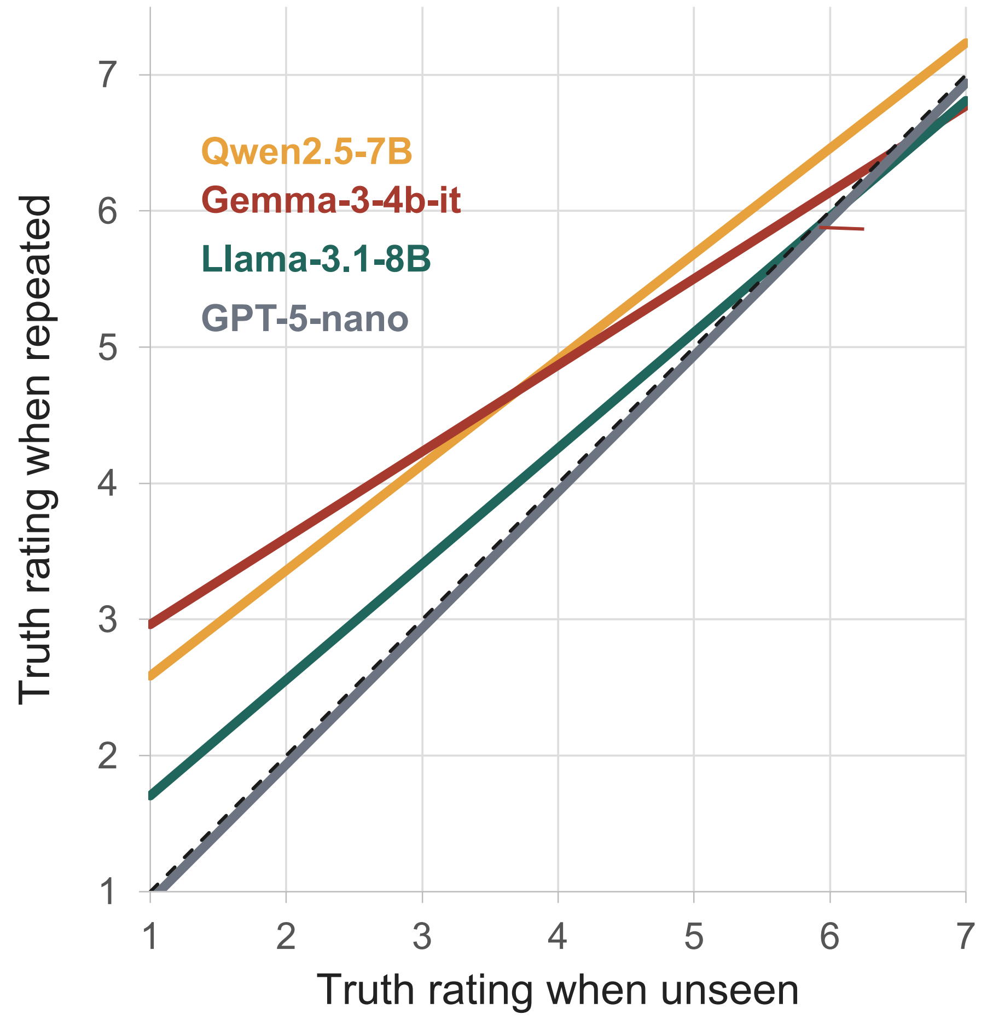
RQ2:
`GPT-5-nano`
`Meta-Llama-3.1-8B-Instruct`
`gemma-3-4b-it`
`Qwen2.5-7B-Instruct`

RQ3: Accounting for random effect from the nested data: the statement and the simulation run (context from news feed)

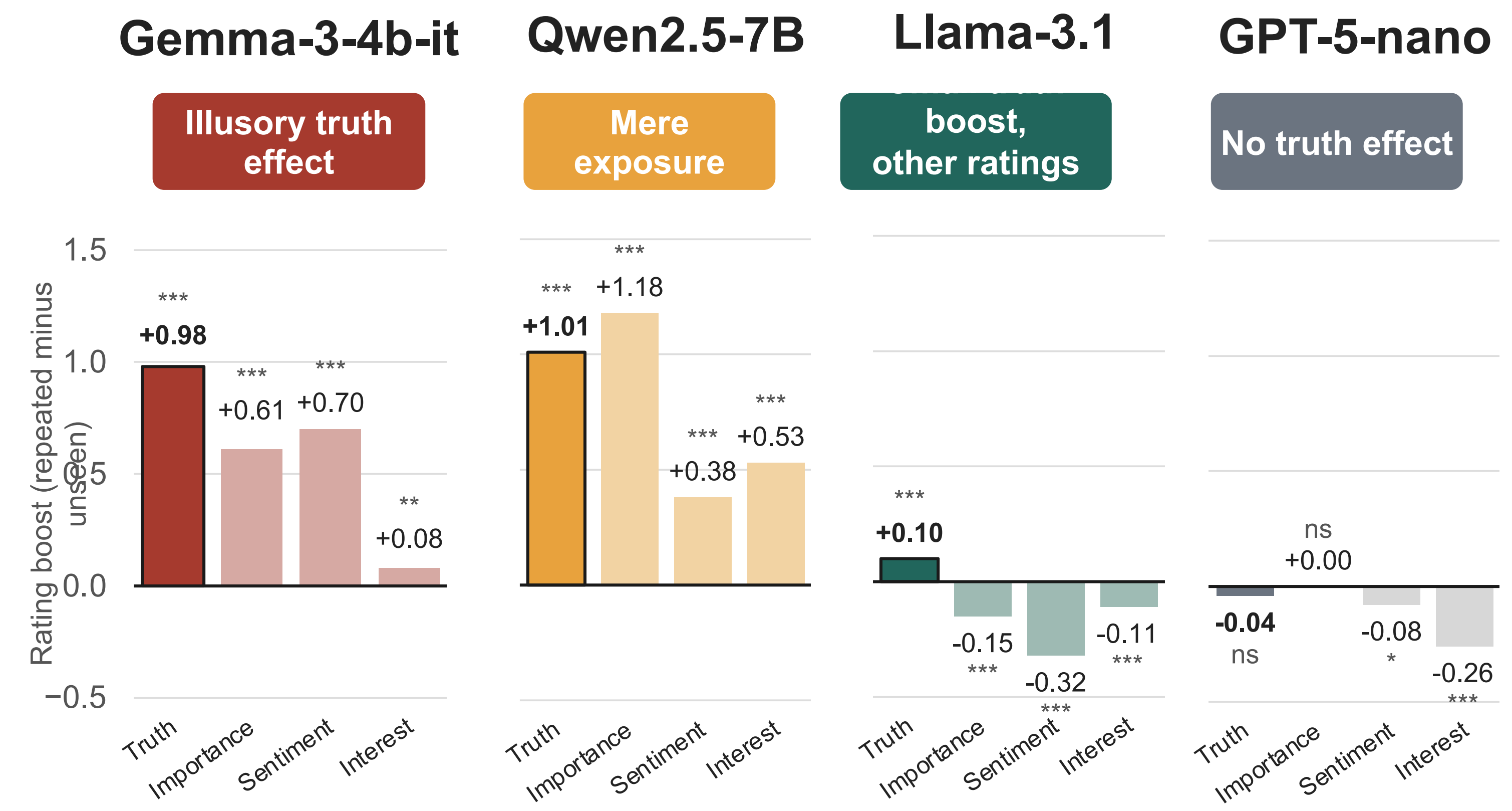
RESULTS



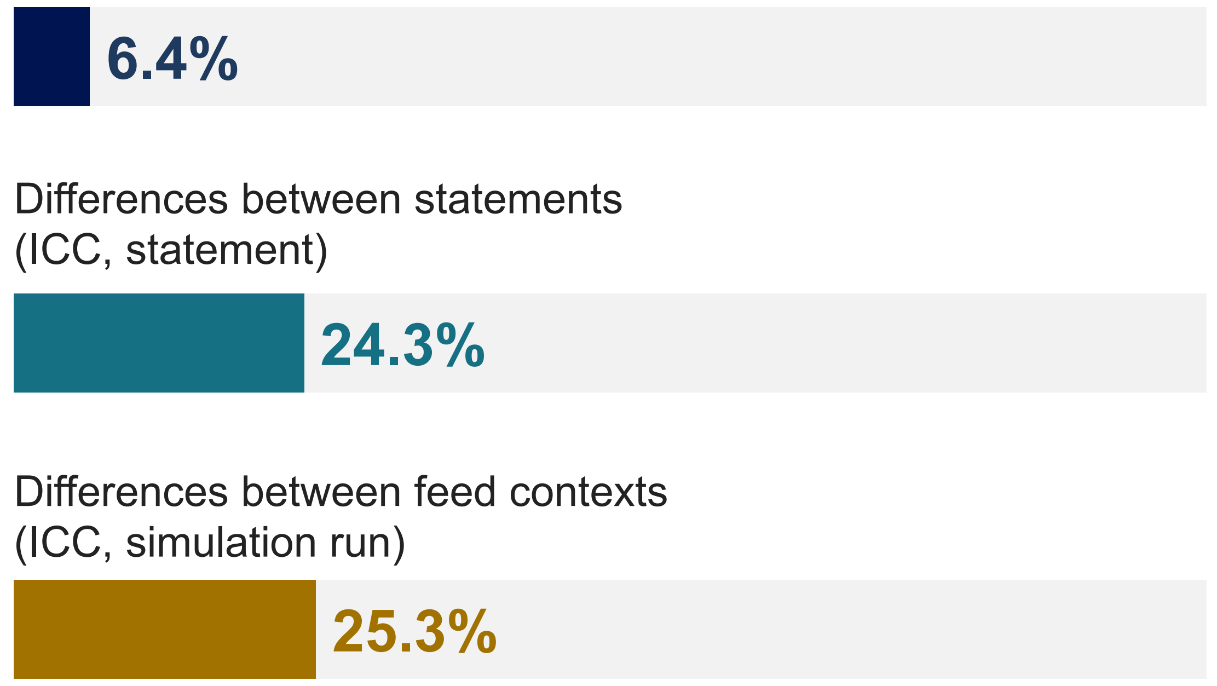
GPT-5-nano runs at a fixed temperature of 1.



OLS fit per model over 100 statements, temperature 1: $r' = r + \text{offset} + (r - 4) \cdot \text{tilt}$. Dashed: no effect.



*Estimated marginal means from the linear mixed-effects model (all temperatures). *** p<.001, ** p<.01, * p<.05, ns not significant.*



Share of rating variance. 0 to 100%.

TAKE-AWAYS

RQ1: ITE in LLMs?	A mixture of ITE, Mere Exposure and No effects.
RQ2: difference in models?	A difference in behavior across LLMs. No difference between temperatures.
RQ3: context?	The effect is real for some models, but highly context sensitive.

Implications for Generative ABM

Gemma-3-4b-it for simulating **misinformation** spread offers the **closest behavioral approximation of ITE** on social media.